adain_summary

Introduction :

- The original Gatys method is highly flexible but relies on an iterative optimization process. However, their frame work requires a slow iterative optimization process, which limits its practical application.
- While feed forward networks improved speed, they were restricted to a fixed, pre defined set of styles and could not adapt to new ones.
- This paper introduces **Adaptive Instance Normalization (AdaIN)**, resolving this flexibility speed dilemma to enable arbitrary style transfer in real time usage by using a single feed forward neural network.

The problem, and its solution :

- The authors realized that standard Instance Normalization (IN) layers perform style normalization by normalizing feature statistics (mean and variance).
- AdaIN takes this forward , by taking a content input (x) and a style input (y), and simply aligning the channel wise mean and variance of the content to match those of the style.
- Mathematically, AdaIN scales the normalized content input with the style's standard deviation ($\sigma(y)$) and shifts it using the style's mean ($\mu(y)$): $AdaIN(x, y) = \sigma(y)\left(\frac{x-\mu(x)}{\sigma(x)}\right) + \mu(y)$.
- Unlike other normalization techniques, AdaIN has no learnable affine parameters; it computes them adaptively directly from the style image.

The Network Architecture

- The system uses a simple encoder-decoder architecture.

- **The Encoder:** The first few layers of a pre-trained VGG-19 network encode the content and style images into the feature space.
- **The AdaIN Layer:** This layer fuses the two by adjusting the content's statistics to match the style's statistics in that feature space.
- **The Decoder:** A randomly initialized network is trained to invert this stylized feature map back into the final image space.
- The decoder does **not** use any Instance Normalization or Batch Normalization layers, as those would incorrectly force the output to normalize toward a single style.

The Result

- The method is nearly three orders of magnitude faster than Gatys' optimization based method, which enables arbitrary style transfer in real time..
- It allows for flexible user controls at runtime, like adjusting the content style trade off, blending multiple styles, and preserving original colors without any modification to the training process.

Reference :

[1703.06868] Arbitrary Style Transfer in Real-time with Adaptive Instance Normalization